

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0175

Monolith 138 kV
October 23, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0175. The Interconnection Customer proposes a 95.3 MW wind facility interconnecting at the Monolith 138 kV substation in the Virginia area. The request was received on November 15, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 111 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,356,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ross Fork – Artesia Country Club 138kV
Contingency: Air Force Base – Artesia Country Club 345kV Line
Loading: 167.6% (Rating: 274 MVA)
DFAX: 9.5%
Pre-project loading: 164.3%
2. Facility: Grimmway-Malaga – Sleepy Hollow Substation 138kV
Contingency: Kearneyville – Clark County 345kV
Loading: 153.3% (Rating: 251 MVA)
DFAX: 7.2%
Pre-project loading: 150.5%
3. Facility: Sleepy Hollow – Millville 138kV
Contingency: Clark County – Kearneyville 345kV
Loading: 152.3% (Rating: 245 MVA)
DFAX: 7.2%
Pre-project loading: 149.5%
4. Facility: Douglas City – Millville 138kV
Contingency: Kearneyville – Clark County 345kV line
Loading: 150.1% (Rating: 242 MVA)
DFAX: 7.2%
Pre-project loading: 147.3%
5. Facility: T.S. Power – Twin Lakes Alte 138kV Line
Contingency: Thomaston – Loudon 345kV
Loading: 150.1% (Rating: 185 MVA)
DFAX: 8.1%
Pre-project loading: 145.9%

6. Facility: G-P Gypsum – Brighton 138kV
Contingency: Artesia Country Club – Air Force Base 345kV
Loading: 138.9% (Rating: 207 MVA)
DFAX: 5.2%
Pre-project loading: 136.5%
7. Facility: Huntley – Twin Lakes Alte 138kV
Contingency: International Paper Valliant Ok – Kearneyville 345kV line
Loading: 136.2% (Rating: 247 MVA)
DFAX: 5.9%
Pre-project loading: 133.9%
8. Facility: Mammoth Pacific 2 – T.S. Power 138kV Line
Contingency: Loudon – Thomaston 345kV Line
Loading: 135.6% (Rating: 191 MVA)
DFAX: 8.1%
Pre-project loading: 131.6%
9. Facility: Killingsworth – Yelm Pp Substation 138kV
Contingency: Celeron – Sandalfort 138kV
Loading: 133.2% (Rating: 238 MVA)
DFAX: 18.8%
Pre-project loading: 125.7%
10. Facility: Shepherdstown Substation – Peachland 138kV
Contingency: Clark County – Kearneyville 345kV
Loading: 128.6% (Rating: 245 MVA)
DFAX: 6.0%
Pre-project loading: 126.3%
11. Facility: Mcconnico – Huntley 138kV L
Contingency: International Paper Valliant Ok – Kearneyville 345kV
Loading: 128.1% (Rating: 267 MVA)
DFAX: 5.9%
Pre-project loading: 126.0%
12. Facility: Peachland – Sun City Substation 138kV
Contingency: Clark County – Kearneyville 345kV
Loading: 127.0% (Rating: 243 MVA)
DFAX: 6.0%
Pre-project loading: 124.7%
13. Facility: Wheeler – Pasco Cnty Solid Waste Resource Recovery 138kV
Contingency: Thoaston – Lodon 345kV
Loading: 125.1% (Rating: 267 MVA)
DFAX: 5.9%
Pre-project loading: 123.0%
14. Facility: Richford – Sun City 138kV
Contingency: Kearneyville – Clark County 345kV
Loading: 124.7% (Rating: 241 MVA)
DFAX: 6.0%
Pre-project loading: 122.3%
15. Facility: University City – Lamont 138kV L
Contingency: Clark County – Kearneyville 345kV
Loading: 122.2% (Rating: 207 MVA)
DFAX: 6.0%
Pre-project loading: 119.4%
16. Facility: University City – Richford 138kV
Contingency: Kearneyville – Clark County 345kV

- Loading: 121.7% (Rating: 235 MVA)
DFAX: 6.0%
Pre-project loading: 119.3%
17. Facility: Sandalfort – North Mesa 138kV
Contingency: Celeron – Yelm Pp 138kV
Loading: 115.6% (Rating: 267 MVA)
DFAX: 19.6%
Pre-project loading: 108.6%
18. Facility: Columbus City – North Mesa 138kV L
Contingency: International Paper Valliant Ok – Kearneyville 345kV
Loading: 115.5% (Rating: 243 MVA)
DFAX: 6.1%
Pre-project loading: 113.1%
19. Facility: Celeron – Yelm Pp 138kV
Contingency: Sandalfort – Celeron 138kV
Loading: 114.6% (Rating: 285 MVA)
DFAX: 18.8%
Pre-project loading: 108.3%
20. Facility: Sandalfort – Celeron 138kV
Contingency: Celeron – Yelm Pp 138kV
Loading: 113.4% (Rating: 283 MVA)
DFAX: 19.6%
Pre-project loading: 106.8%
21. Facility: Gallegos – Killingsworth 138kV
Contingency: Kearneyville – Clark Conty Substation 345kV
Loading: 112.3% (Rating: 257 MVA)
DFAX: 8.2%
Pre-project loading: 109.2%
22. Facility: Gallegos – Redman 138kV
Contingency: Clark County – Kearneyville 345kV
Loading: 110.8% (Rating: 253 MVA)
DFAX: 8.2%
Pre-project loading: 107.7%
23. Facility: Glenbrook Substation – Redman 138kV Line
Contingency: Clark County – Kearneyville 345kV Line
Loading: 109.9% (Rating: 249 MVA)
DFAX: 8.2%
Pre-project loading: 106.8%
24. Facility: Mammoth Pacific 2 – Wheeler Substation 138kV
Contingency: Loudon – Thomaston 345kV Line
Loading: 108.4% (Rating: 215 MVA)
DFAX: 8.1%
Pre-project loading: 104.8%
25. Facility: Glenbrook – Palmyra 138kV line
Contingency: Clark County – Kearneyville 345kV
Loading: 107.9% (Rating: 244 MVA)
DFAX: 8.2%
Pre-project loading: 104.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ross Fork – Artesia Country Club Substation 138kV	Rebuild line with higher-capacity conductor	\$4,567,000	18% (\$826,000)
Sleepy Hollow – Grimway-Malaga 138kV	Rebuild line with higher-capacity conductor	\$2,379,000	14% (\$337,000)
Millville – Sleepy Hollow 138kV line	Rebuild line with higher-capacity conductor	\$9,175,000	14% (\$1,301,000)
Douglas City – Millville 138kV	Rebuild line with higher-capacity conductor	\$8,284,000	14% (\$1,175,000)
T.S. Power – Twin Lakes Alte 138kV Line	Rebuild line with higher-capacity conductor	\$2,618,000	8% (\$211,000)
G-P Gypsum – Brighton 138kV	Rebuild line with higher-capacity conductor	\$6,752,000	33% (\$2,253,000)
Huntley – Twin Lakes Lte 138kV	Rebuild line with higher-capacity conductor	\$17,605,000	7% (\$1,229,000)
Mammoth Pacific 2 – T.S. Power 138kV Line	Rebuild line with higher-capacity conductor	\$4,617,000	8% (\$372,000)
Killingsworth – Yelm Pp Substation 138kV	Rebuild line with higher-capacity conductor	\$1,551,000	17% (\$263,000)
Shepherdstown Substation – Peachland 138kV	Rebuild line with higher-capacity conductor	\$1,650,000	25% (\$417,000)
Mcconnico Substation – Huntley 138kV line	Rebuild line with higher-capacity conductor	\$6,529,000	7% (\$456,000)
Peachland – Sun City 138kV	Rebuild line with higher-capacity conductor	\$3,475,000	25% (\$879,000)
Pasco Cnty Solid Waste Resource Recovery – Wheeler 138kV	Rebuild line with higher-capacity conductor	\$4,703,000	8% (\$376,000)
Richford – Sun City 138kV Line	Rebuild line with higher-capacity conductor	\$13,702,000	25% (\$3,463,000)
University City – Lamont 138kV L	Rebuild line with higher-capacity conductor	\$13,140,000	25% (\$3,322,000)
University City – Richford 138kV	Rebuild line with higher-capacity conductor	\$5,265,000	25% (\$1,331,000)
Sandalfoot – North Mesa 138kV	Rebuild line with higher-capacity conductor	\$1,463,000	17% (\$248,000)
Columbus City – N Mesa 138kV	Rebuild line with higher-capacity conductor	\$1,499,000	20% (\$303,000)
Celeron – Yelm Pp 138kV	Rebuild line with higher-capacity conductor	\$7,391,000	17% (\$1,251,000)
Celeron – Sandalfoot 138kV line	Rebuild line with higher-capacity conductor	\$3,271,000	17% (\$554,000)
Gallegos – Killingsworth 138kV	Rebuild line with higher-capacity conductor	\$3,885,000	22% (\$860,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gallegos – Redman 138kV	Rebuild line with higher-capacity conductor	\$1,822,000	22% (\$403,000)
Glenbrook Substation – Redman 138kV Line	Reconductor line	\$893,000	22% (\$198,000)
Mammoth Pacific 2 – Wheeler Substation 138kV	Reconductor line	\$2,265,000	8% (\$182,000)
Glenbrook – Palmyra 138kV line	Reconductor line	\$658,000	22% (\$146,000)
	Total allocated to project		\$22,356,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.